

BASIC ELETRONICS

LECTURE 3:

BJT BIASING CIRCUITS

OUTLINE

- Introduction
- Application
- The DC Operating Point
- Voltage-Divider Bias



INTRODUCTION

- As we have learned in **lecture 2**, a transistor must be properly biased in order to operate as an **amplifier**.
- **Transistor Biasing** is the process of setting a transistors **DC operating voltage** or **current conditions** to the correct level so that any AC input signal can be amplified correctly by the transistor.



INTRODUCTION

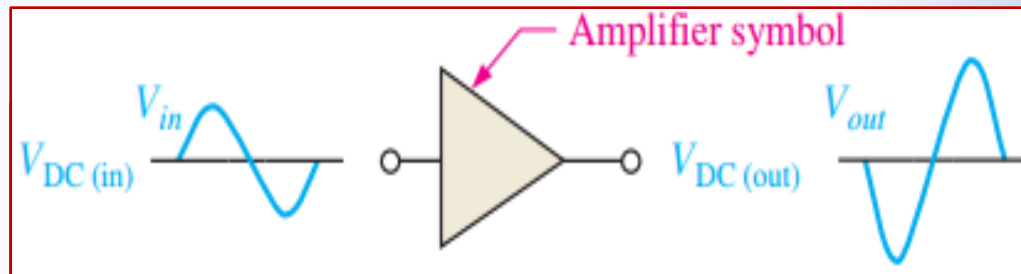
- DC biasing is used to establish fixed dc values for the transistor currents and voltages called the dc operating point or quiescent point (Q-point).
- A dc operating point must be set so that signal variations at the input terminal are amplified and accurately reproduced at the output terminal.
- This means that, at the dc operating point, I_C and V_{CE} have specified values.
- In this lecture, several types of bias circuits will be discussed.



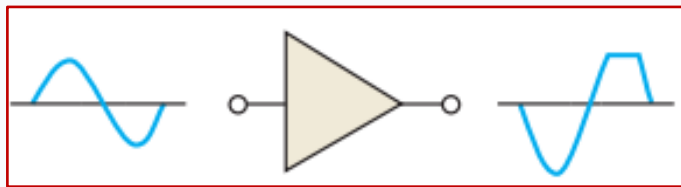
THE DC OPERATING POINT

DC Bias

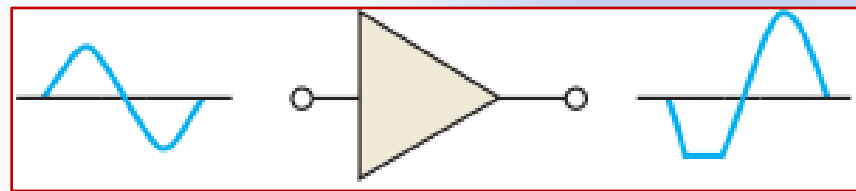
- Bias establishes the dc **operating point** (Q-point) for proper linear operation of an amplifier.
- If an amplifier is not biased with correct dc voltages on the input and output, it can go into **saturation** or **cutoff** when an input signal is applied.



- **Linear operation:** larger output has same shape as input except that it is inverted



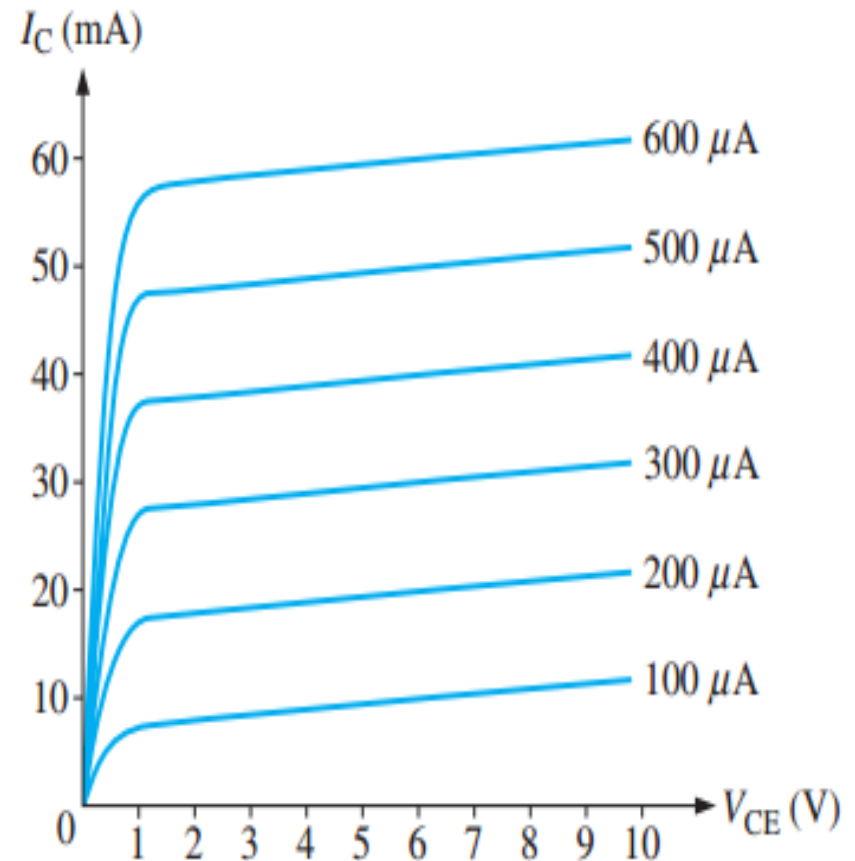
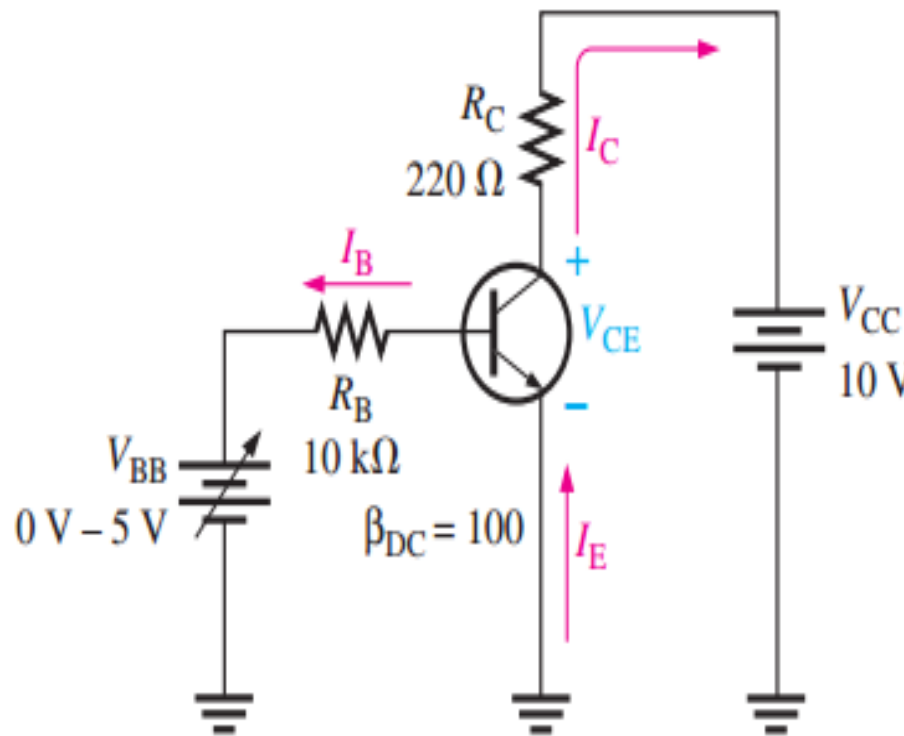
- **Nonlinear operation:** output voltage limited (clipped) by **cutoff**



- **Nonlinear operation:** output voltage limited (clipped) by **saturation**

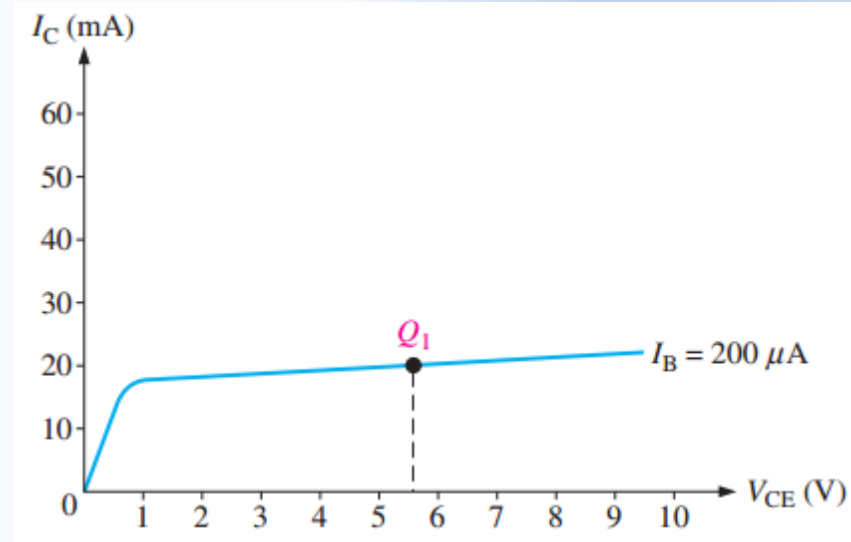
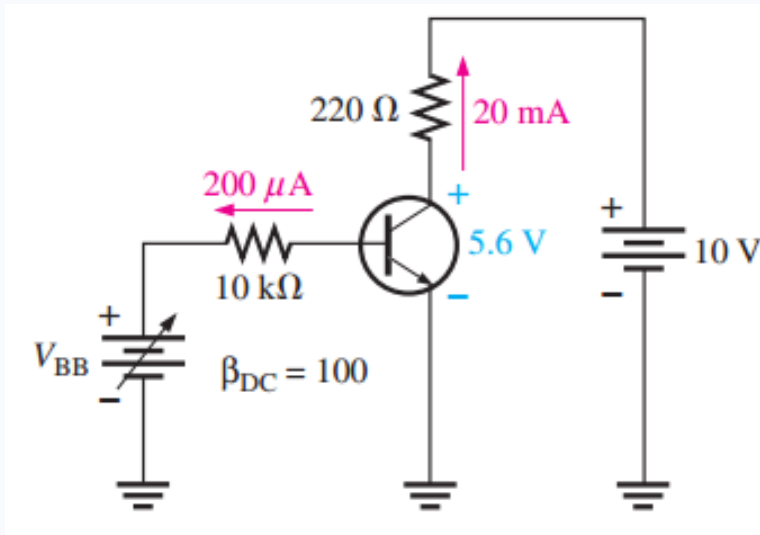
THE DC OPERATING POINT

- Consider a dc-biased transistor circuit with variable bias voltage (V_{BB}) for generating the collector characteristic curves.



THE DC OPERATING POINT

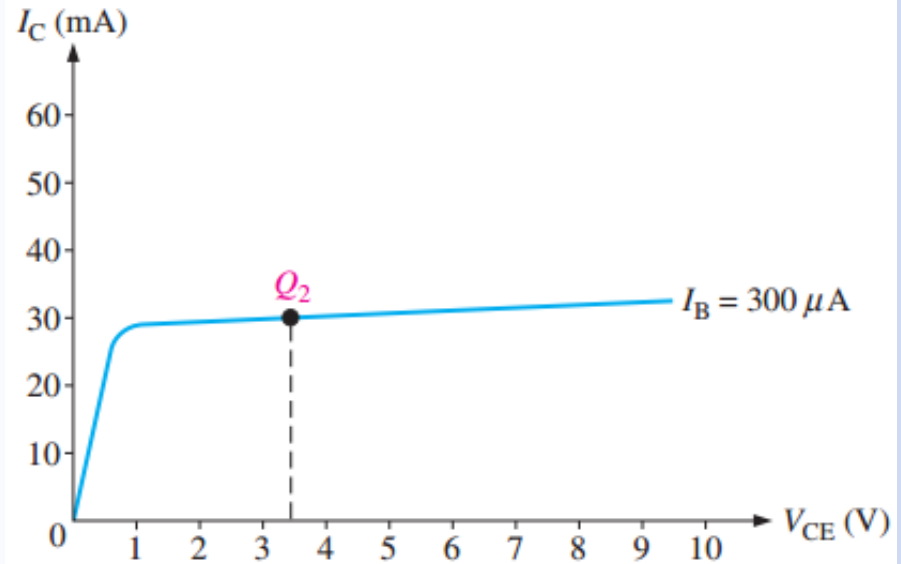
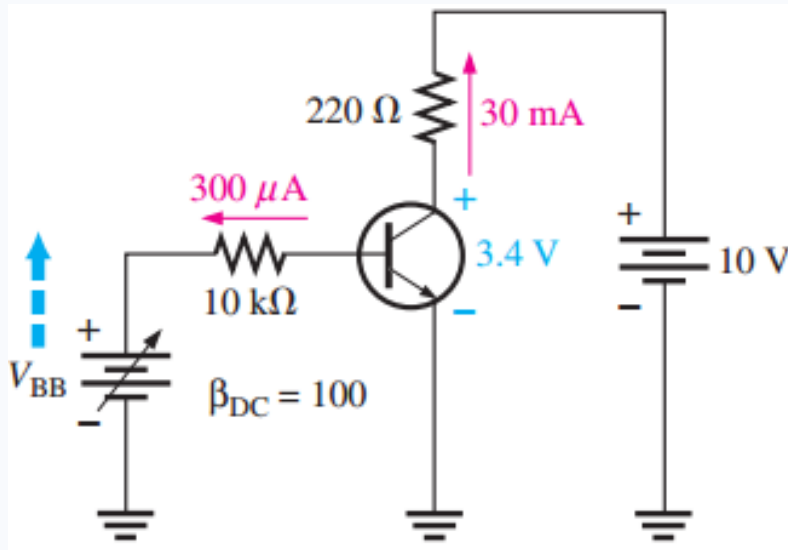
- First, V_{BB} is adjusted to produce an I_B of $200\ \mu\text{A}$.
- Since $I_C = \beta_{DC} I_B$ the collector current is **20 mA**



- $V_{CE} = V_{CC} - I_C R_C = 10\text{ V} - 20\text{ mA} (220\ \Omega) = 5.6\text{ V}$
- The Q-point Q_1 is $(V_{CE}, I_C) = (5.6\text{ V}, 20\text{ mA})$

THE DC OPERATING POINT

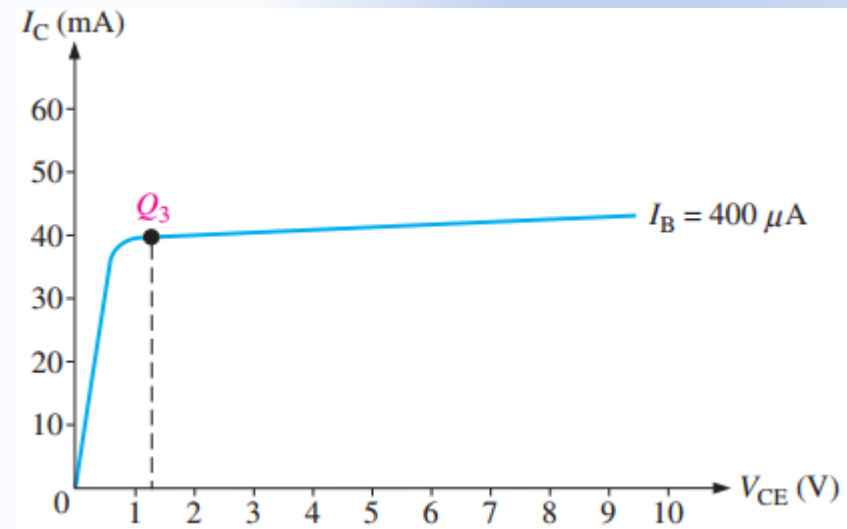
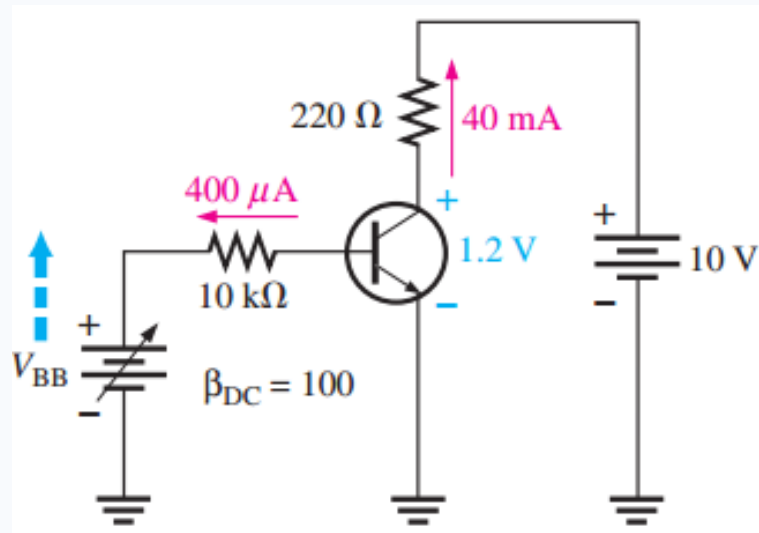
- Next, V_{BB} is increased to produce an I_B of $300\ \mu\text{A}$ and an I_C of $30\ \text{mA}$.



- $V_{CE} = V_{CC} - I_C R_C = 10\text{V} - 30\text{mA} (220\Omega) = 3.4\text{V}$
- The Q-point Q_1 is $(V_{CE}, I_C) = (3.4\text{V}, 30\text{mA})$

THE DC OPERATING POINT

- Finally, V_{BB} is increased to give an I_B of $400\ \mu\text{A}$ and an I_C of $40\ \text{mA}$

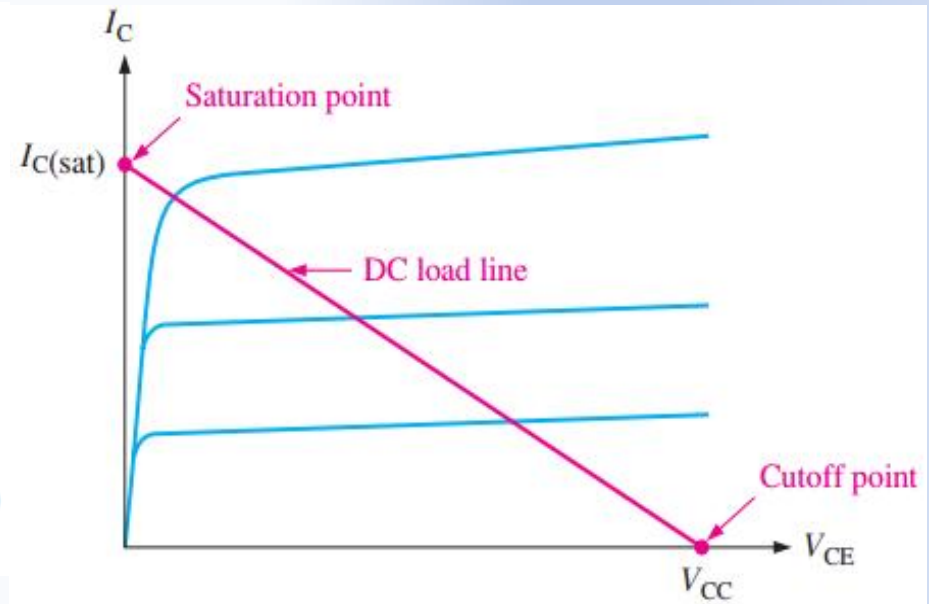
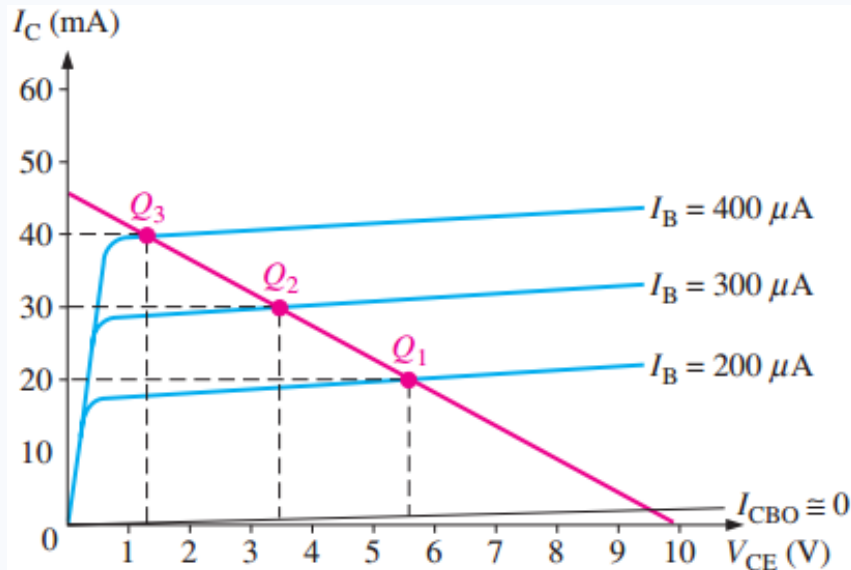


- $V_{CE} = V_{CC} - I_C R_C = 10\text{V} - 40\text{mA} (220\Omega) = 1.2\text{V}$
- The Q-point Q_1 is $(V_{CE}, I_C) = (1.2\text{V}, 30\text{mA})$

THE DC OPERATING POINT

DC Load Line

- This is a straight line drawn on the characteristic curves from the saturation value where $I_C = I_{C(sat)}$ on the y-axis to the cutoff value where $V_{CE} = V_{CC}$ on the x-axis



THE DC OPERATING POINT

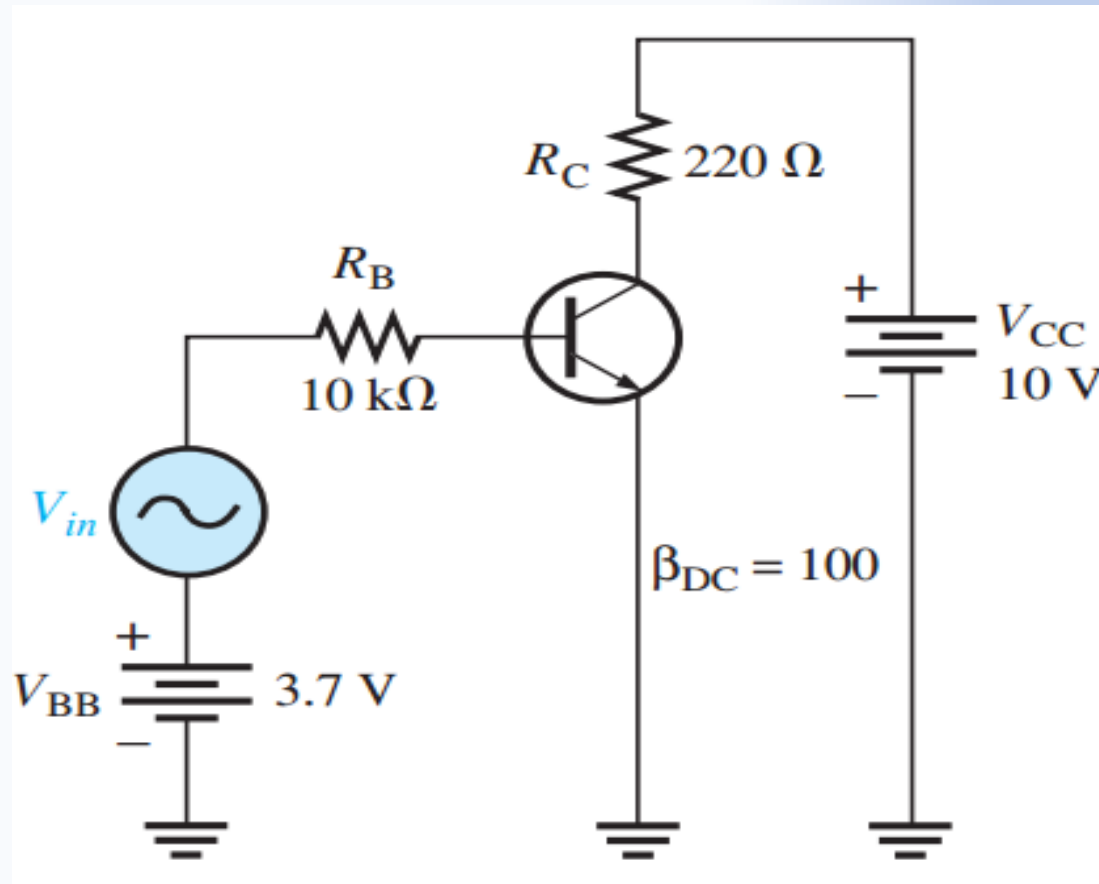
Linear Operation

- The region along the load line including all points between saturation and cutoff is generally known as the **linear region** of the transistor's operation.
- As long as the transistor is operated in this region, the output voltage is ideally a **linear reproduction of the input**.



THE DC OPERATING POINT

Linear Operation



THE DC OPERATING POINT

Linear Operation

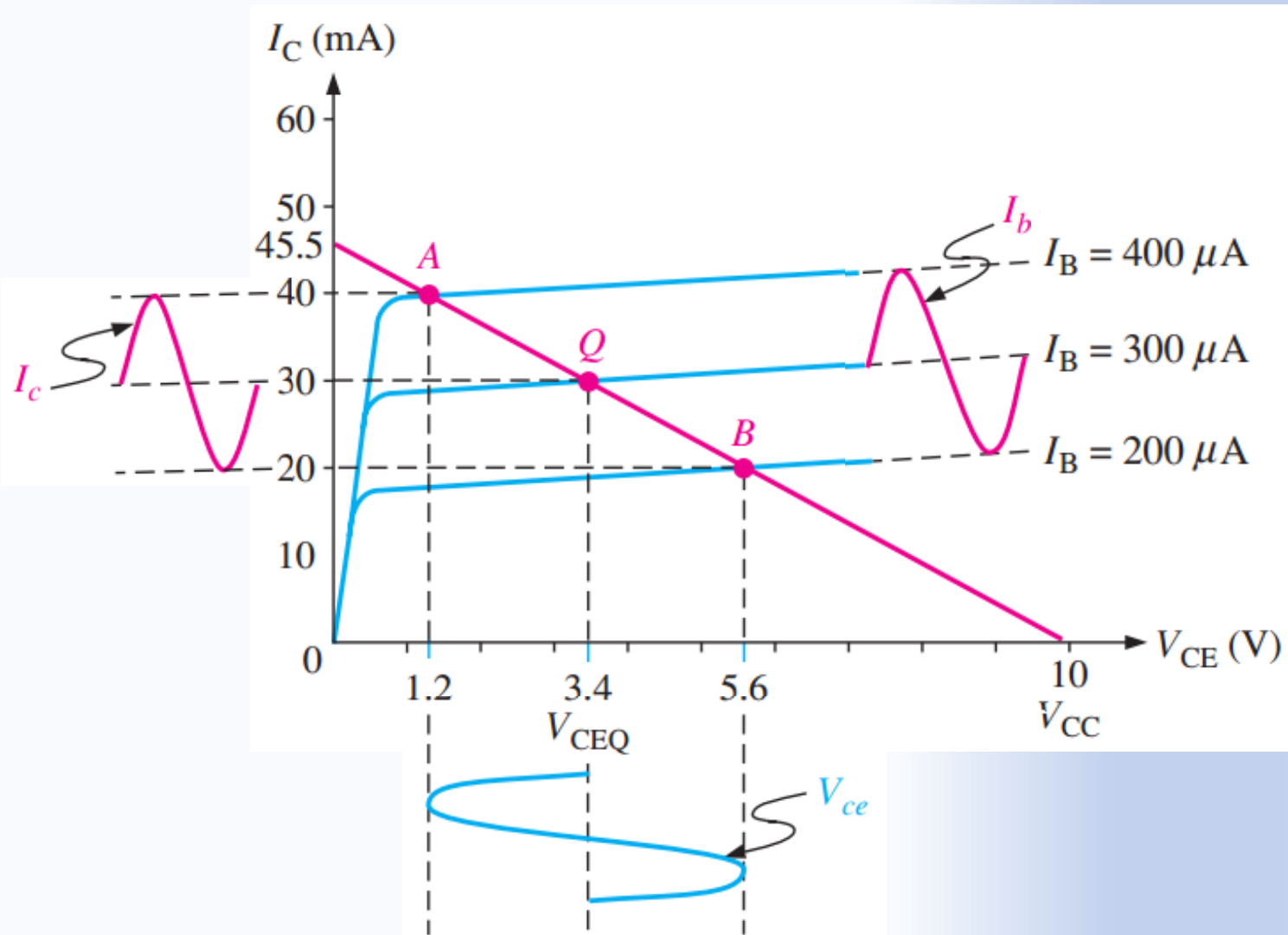
$$I_{BQ} = \frac{V_{BB} - 0.7 \text{ V}}{R_B} = \frac{3.7 \text{ V} - 0.7 \text{ V}}{10 \text{ k}\Omega} = 300 \mu\text{A}$$

$$I_{CQ} = \beta_{DC} I_{BQ} = (100)(300 \mu\text{A}) = 30 \text{ mA}$$

$$V_{CEQ} = V_{CC} - I_{CQ} R_C = 10 \text{ V} - (30 \text{ mA})(220 \Omega) = 3.4 \text{ V}$$

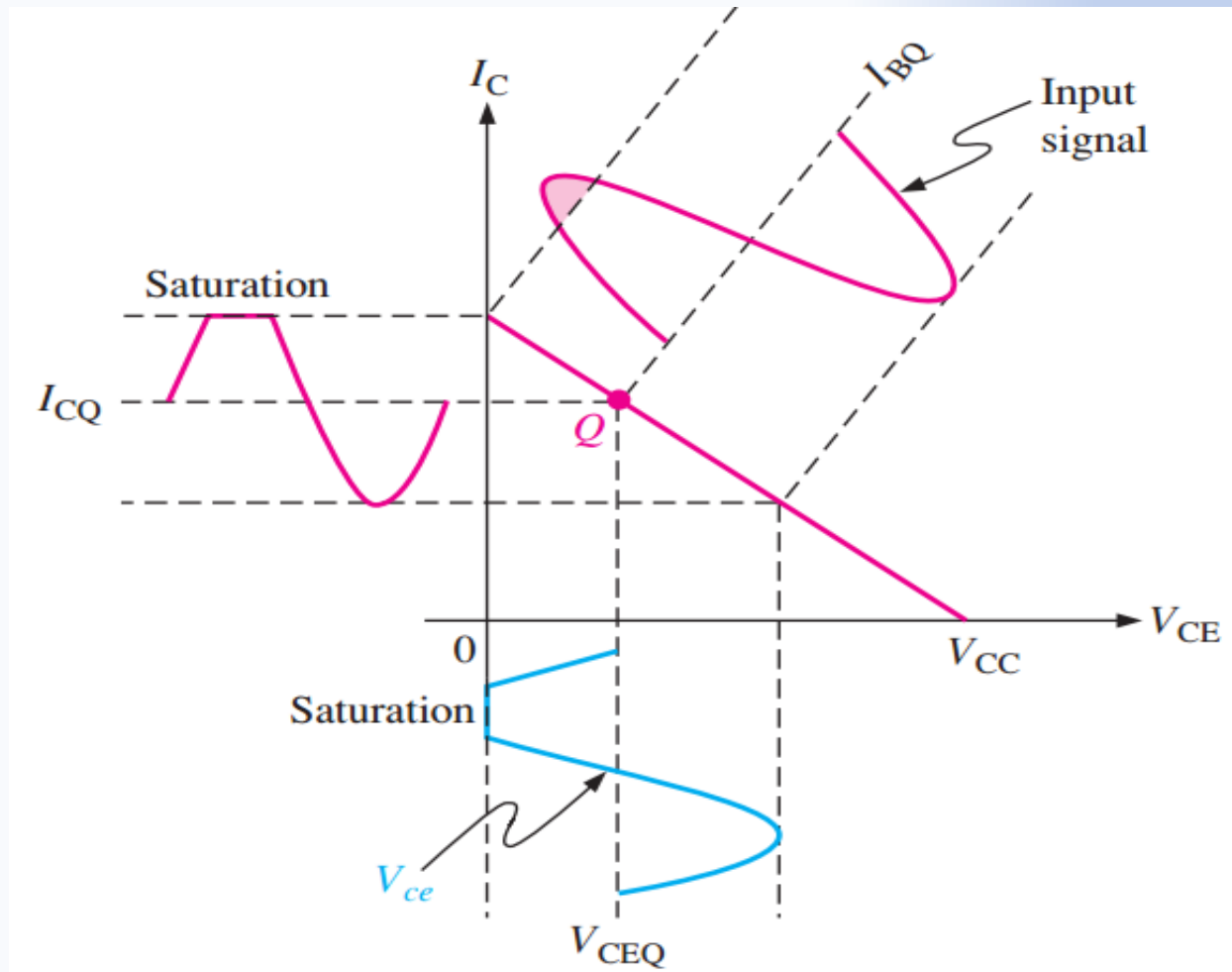


THE DC OPERATING POINT



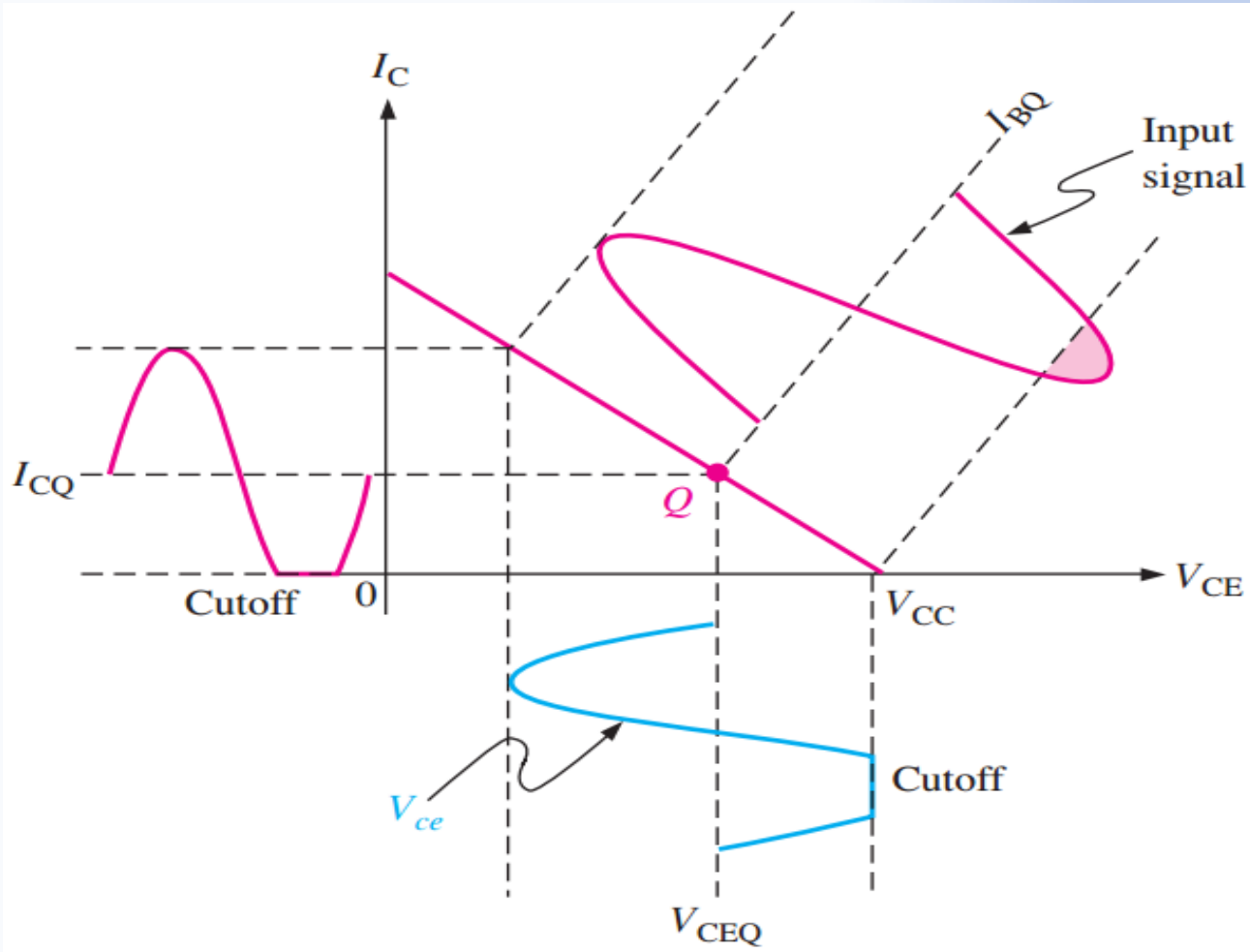
THE DC OPERATING POINT

Waveform Distortion



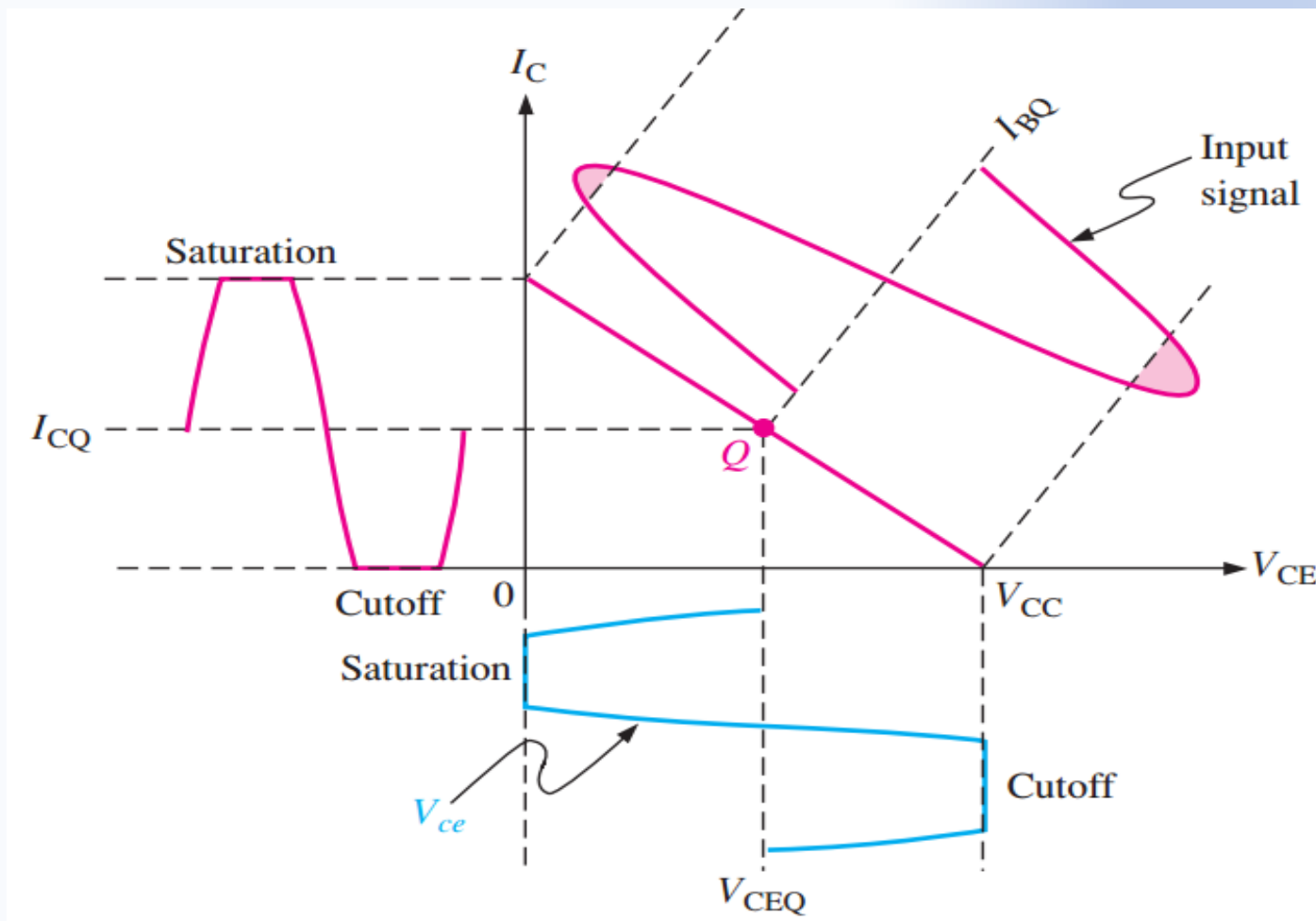
THE DC OPERATING POINT

Waveform Distortion



THE DC OPERATING POINT

Waveform Distortion



THE DC OPERATING POINT

- From a dc-biased transistor circuit;

$$V_{CC} = I_C R_C + V_{CE}$$

- Now the equation for I_C is;

$$I_C = \frac{V_{CC} - V_{CE}}{R_C} = \frac{V_{CC}}{R_C} - \frac{V_{CE}}{R_C}$$
$$I_C = -\frac{V_{CE}}{R_C} + \frac{V_{CC}}{R_C} = -\left(\frac{1}{R_C}\right)V_{CE} + \frac{V_{CC}}{R_C}$$
$$I_C = -\left(\frac{1}{R_C}\right)V_{CE} + \frac{V_{CC}}{R_C}$$

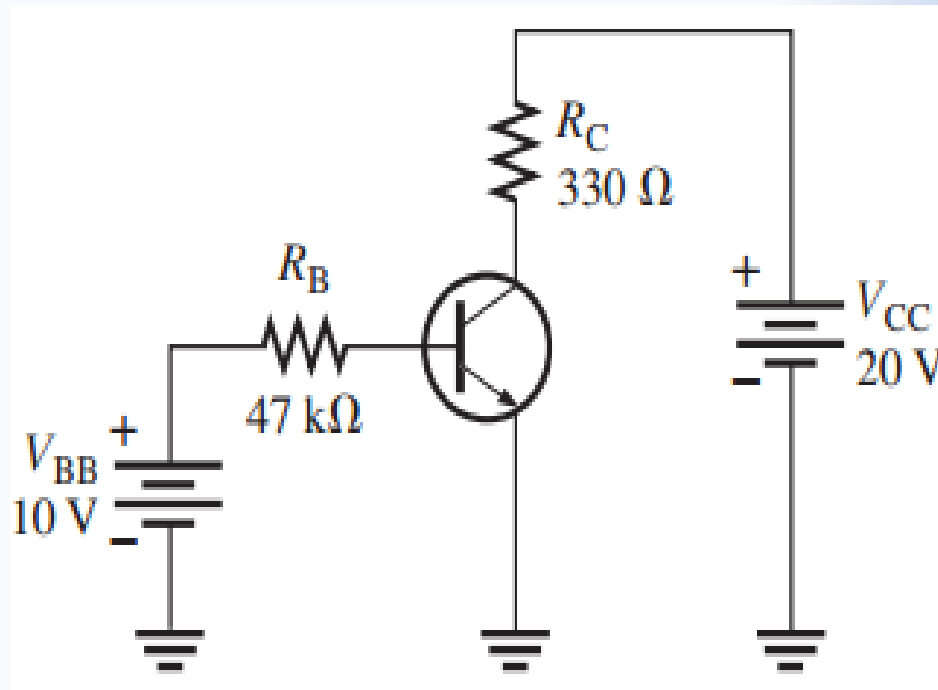
- This is the equation of a straight line with a slope of $-1/R_C$, an x intercept of $V_{CE} = V_{CC}$, and a y intercept of V_{CC}/R_C , which is $I_{C(sat)}$



THE DC OPERATING POINT

Example 1

- Determine the Q-point for the circuit in figure and draw the dc load line. Assume $\beta_{DC} = 200$



THE DC OPERATING POINT

Solution

- The Q-point is defined by the values of I_C and V_{CE} .

$$I_B = \frac{V_{BB} - V_{BE}}{R_B} = \frac{10V - 0.7V}{47 k\Omega} = 198 \mu A$$

$$I_C = \beta_{DC} I_B = 200(198 \mu A) = \mathbf{39.6 mA}$$

$$V_{CE} = V_{CC} - I_C R_C = 10 V - 13.7 V = \mathbf{6.93V}$$

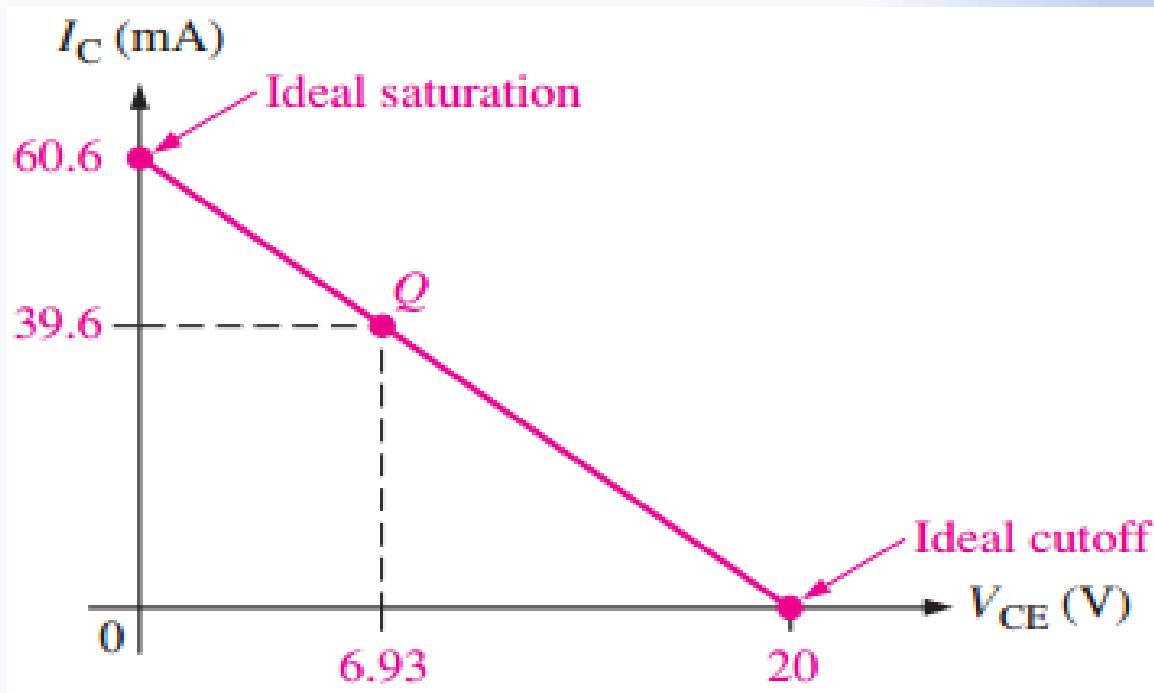
- The Q-point is at $I_C = 39.6 mA$ and at $V_{CE} = 6.93 V$.



THE DC OPERATING POINT

- Since at cutoff; $I_{C(cutoff)} = 0$, then $V_{CE} = V_{CC} = 20\text{ V}$
- Now at saturation; $I_{C(sat)}$ can be calculated as;

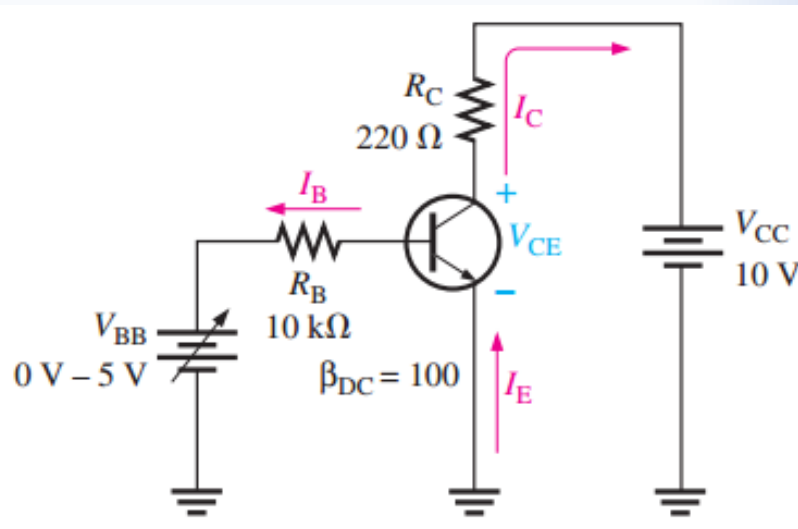
$$I_{C(sat)} = \frac{V_{CC}}{R_C} = \frac{20\text{ V}}{300\ \Omega} = 60.6\text{ mA}$$



THE DC OPERATING POINT

Review question

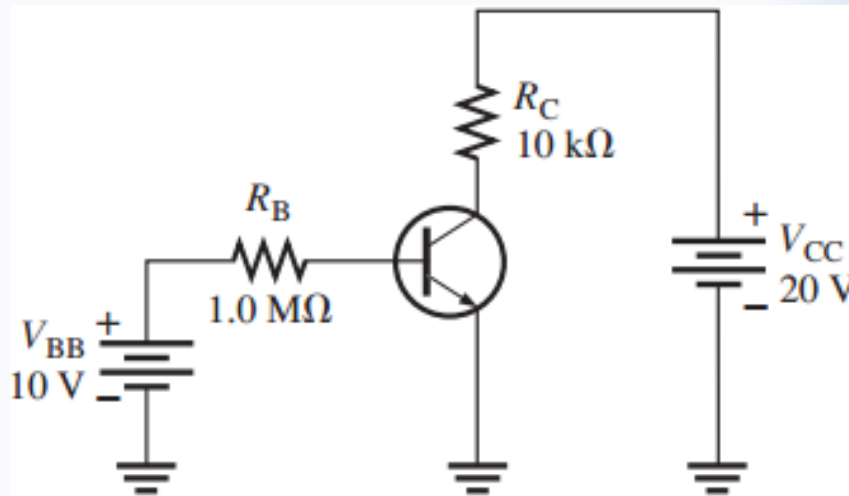
1. Determine the Q-point for the circuit in figure of **example 1** and draw the dc load line for the following circuit values: $\beta_{DC} = 100$, $R_C = 1\text{ k}\Omega$ and $V_{CC} = 24\text{ V}$.
2. What is the Q-point for a biased transistor as in Figure below. $I_B = 150\mu\text{A}$, $\beta_{DC} = 75$, $V_{CC} = 18\text{ V}$ and $R_C = 1\text{ k}\Omega$



THE DC OPERATING POINT

Review question

3. What is the saturation value of collector current and the cutoff value of V_{CE} in **Problem 2**? Draw dc load line.
4. Determine the intercept points of the dc load line on the vertical and horizontal axes of the collector-characteristic curves for the circuit in Figure below.



THE DC OPERATING POINT

Review question

5. Assume that you wish to bias the transistor in Figure of **problem 4** with $I_B = 20 \mu A$. To what voltage must you change the V_{BB} supply? What are I_C and V_{CE} at the Q-point, given that $\beta_{DC} = 50$?

6. From the collector characteristic curves and the dc load line in Figure given in next page, determine the following:

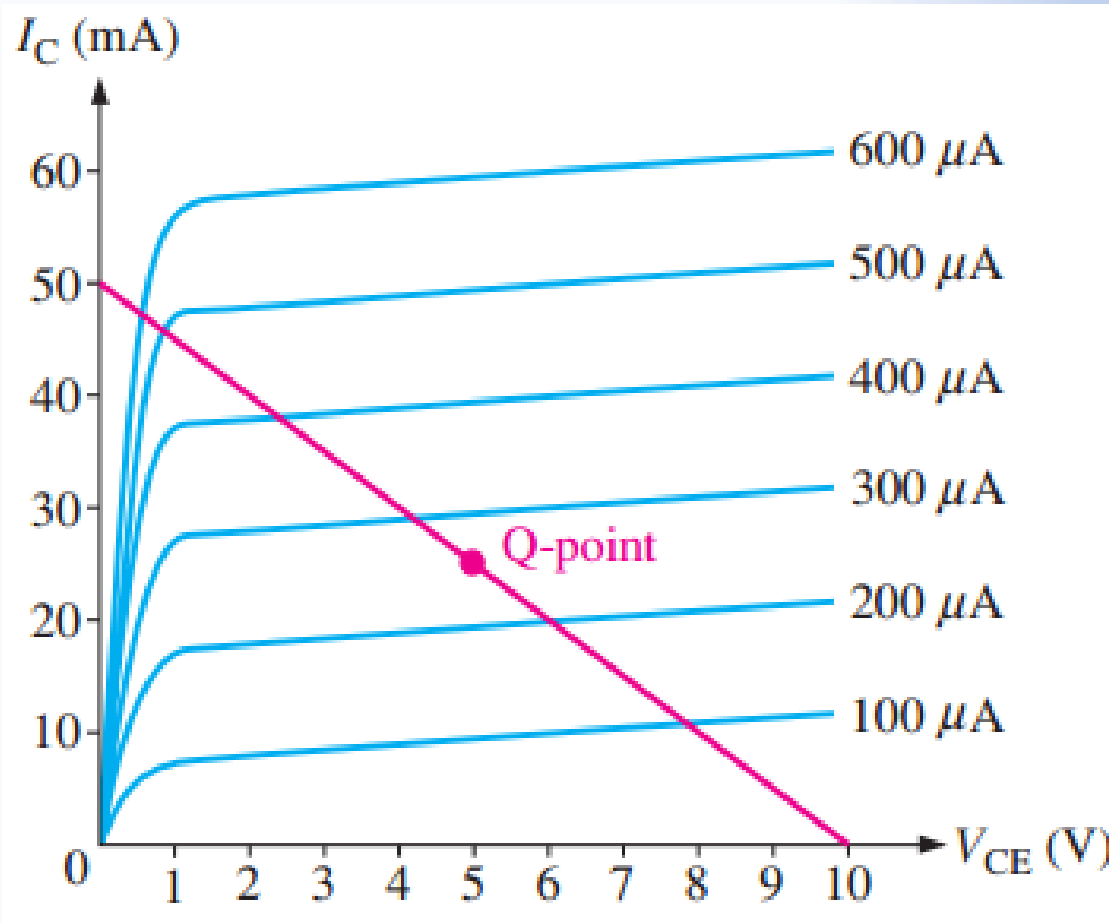
- (a) Collector saturation current
- (b) V_{CE} at cutoff
- (c) Q-point values of I_B , I_C , and V_{CE}



THE DC OPERATING POINT

Review question

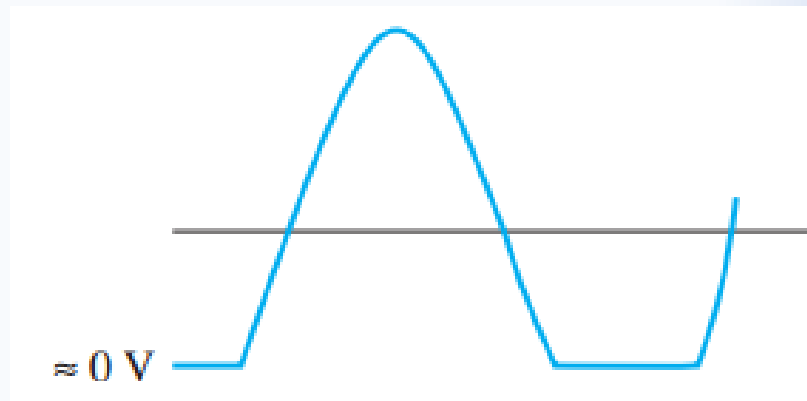
6.



THE DC OPERATING POINT

Review question

7. (i) At what point on the load line does saturation occur? (ii) At what point does cutoff occur?
8. Define Q-point
9. The output (collector voltage) of a biased transistor amplifier is shown in the Figure below. Is the transistor biased too close to cutoff or too close to saturation?

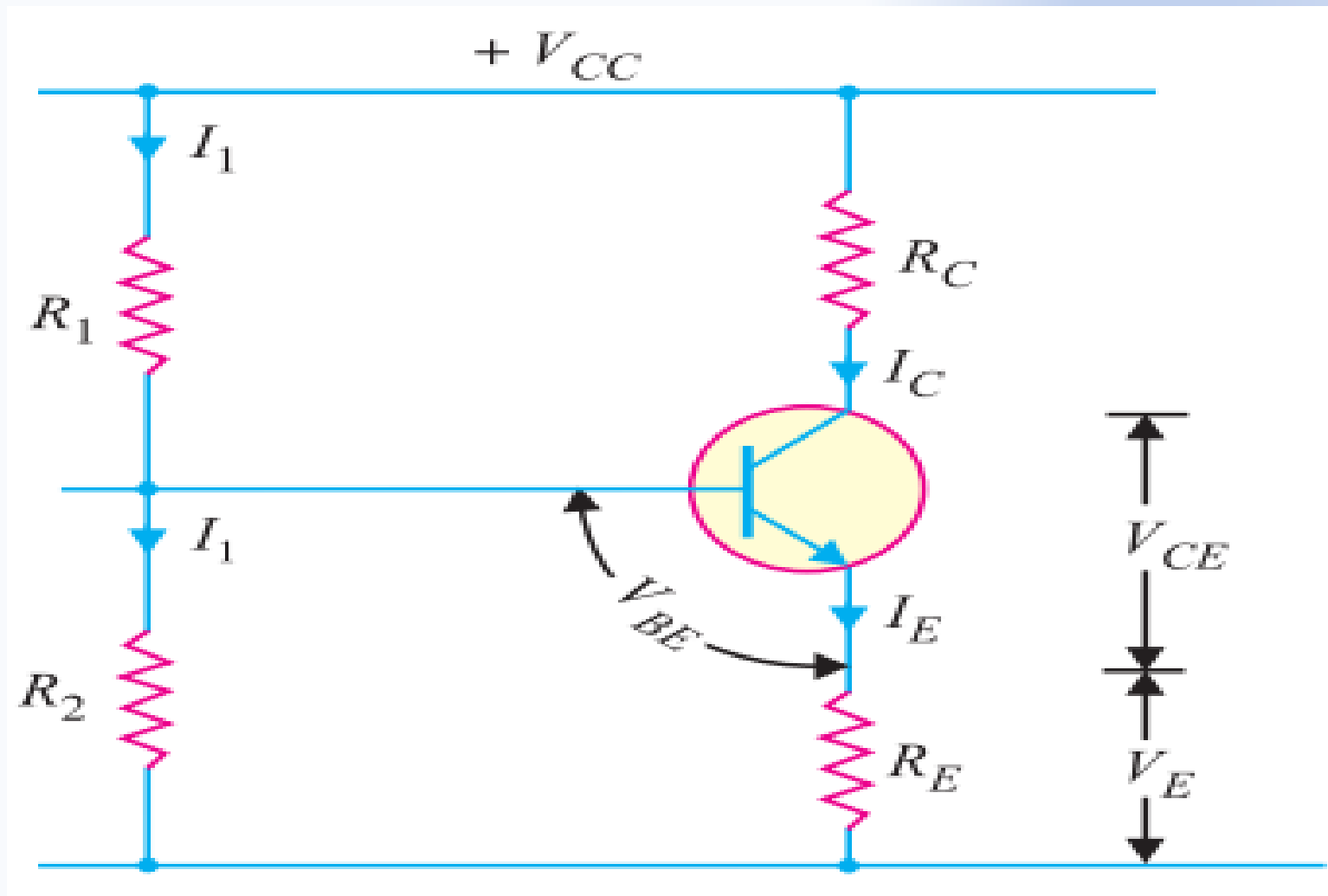


VOLTAGE-DIVIDER BIAS

- In this method, two resistances R_1 and R_2 are connected across the supply voltage V_{CC} and provide biasing.
- The emitter resistance R_E provides **stabilization**.
- The name “voltage divider” comes from the voltage divider formed by R_1 and R_2 .
- The voltage drop across R_2 forward biases the base-emitter junction.
- This causes the base current and hence collector current flow in the zero signal conditions.



VOLTAGE-DIVIDER BIAS



VOLTAGE-DIVIDER BIAS

Collector current I_C

- Suppose that the current flowing through resistance R_1 is I_1 .
- As base current I_B is very small, therefore, it can be assumed with reasonable accuracy that current flowing through R_2 is also I_1 .

$$I_1 = \frac{V_{CC}}{R_1 + R_2}$$

∴ Voltage across resistance R_2 is

$$V_2 = \left(\frac{V_{CC}}{R_1 + R_2} \right) R_2$$



VOLTAGE-DIVIDER BIAS

- Applying Kirchhoff 's voltage law to the base circuit;

$$V_2 = V_{BE} + V_E$$

$$V_2 = V_{BE} + I_E R_E$$

$$I_E = \frac{V_2 - V_{BE}}{R_E}$$

Since $I_C \cong I_E$

$$I_C = \frac{V_2 - V_{BE}}{R_E}$$



VOLTAGE-DIVIDER BIAS

- It is clear from expression that I_C does not at all depend upon β .
- Though I_C depends upon V_{BE} but in practice $V_2 \gg V_{BE}$ so that I_C is practically independent of V_{BE} .
- Thus, I_C in this circuit is almost independent of transistor parameters and hence **good stabilization** is ensured.
- It is due to this reason that potential divider bias has become universal method for providing transistor biasing.



VOLTAGE-DIVIDER BIAS

Collector-emitter voltage V_{CE}

- Applying Kirchhoff 's voltage law to the collector side,

$$V_{CC} = I_C R_C + V_{CE} + I_E R_E$$

Since $I_C \cong I_E$

$$V_{CC} = I_C R_C + V_{CE} + I_C R_E$$

$$V_{CC} = I_C (R_C + R_E) + V_{CE}$$

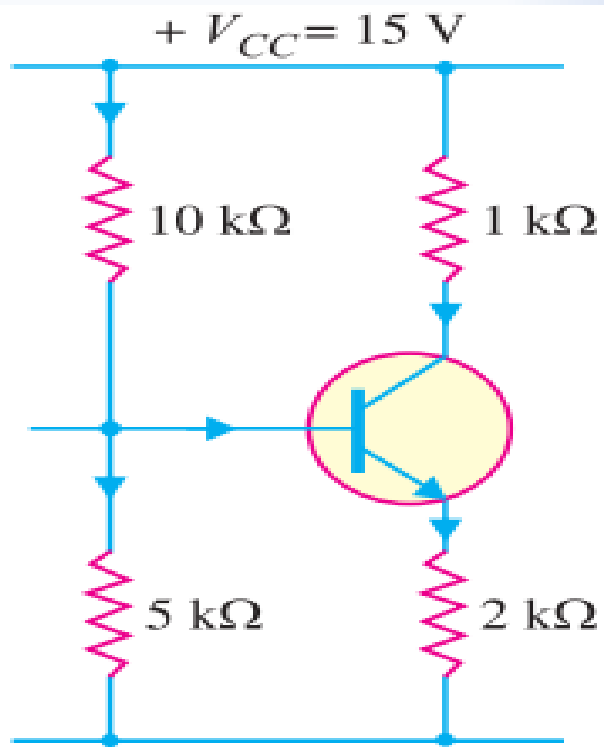
$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$



VOLTAGE-DIVIDER BIAS

Example 2

- The figure below shows the voltage divider bias method. Draw the d.c load line and determine the operating point. Assume the transistor to be of silicon.



VOLTAGE-DIVIDER BIAS

Solution

d.c load line

- The collector-emitter voltage V_{CE} is given by :

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

- When $I_C = 0$,

$$V_{CE} = V_{CC} = 15 \text{ V.}$$

- When $V_{CE} = 0$,

$$I_C = \frac{V_{CC}}{R_C + R_E} = \frac{15 \text{ V}}{(1 + 2) \text{ k}\Omega} = 5 \text{ mA}$$



VOLTAGE-DIVIDER BIAS

Solution

Operating point

- For silicon transistor,

$$V_{BE} = 0.7 \text{ V}$$

- Voltage across $5 \text{ k}\Omega$ is;

$$V_2 = \left(\frac{V_{CC}}{R_1 + R_2} \right) R_2 = \left(\frac{15}{10 + 5} \right) 5 = 5 \text{ V}$$

$\therefore$ Emitter current,

$$I_E = \frac{V_2 - V_{BE}}{R_E} = \frac{5 - 0.7}{2 \text{ k}} = \frac{4.3}{2 \text{ k}} = 2.15 \text{ mA}$$

$\therefore$ Collector current is $I_C \cong I_E = 2.15 \text{ mA}$



VOLTAGE-DIVIDER BIAS

Solution

Collector-emitter voltage,

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

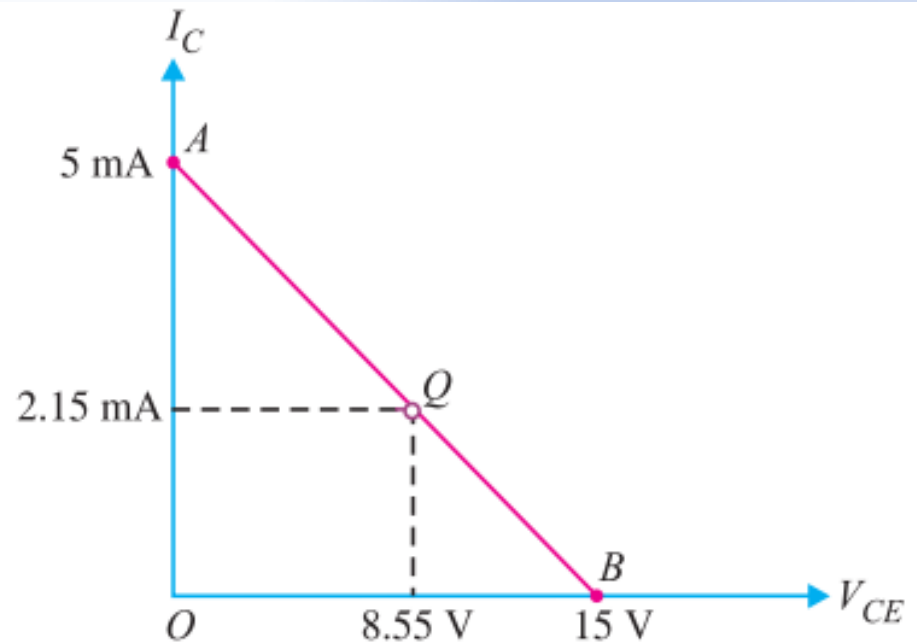
$$V_{CE} = 15\text{ V} - 2.15\text{ mA}(3\text{ k}\Omega)$$

$$V_{CE} = 15\text{ V} - 6.65\text{ V}$$

$$V_{CE} = 8.55\text{ V}$$

∴ Operating point is

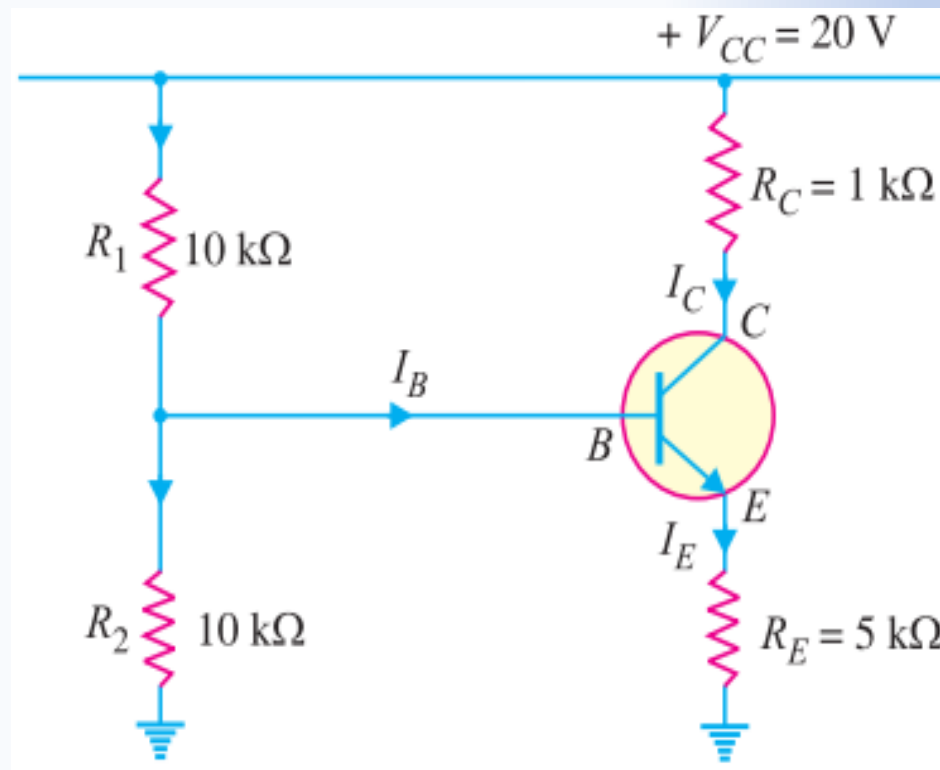
8.55 V, 2.15 mA.



VOLTAGE-DIVIDER BIAS

Review question

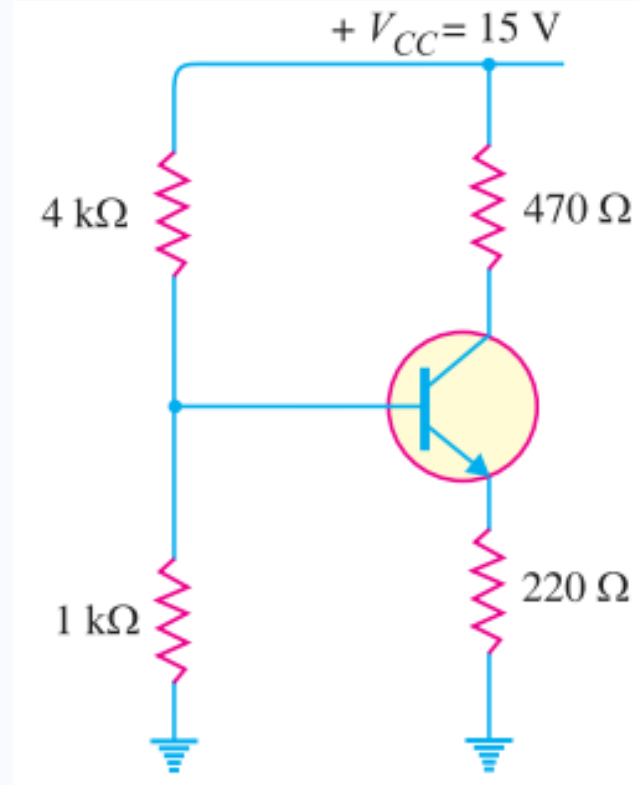
1. Calculate the emitter current in the voltage divider circuit shown in the below. Also find the value of V_{CE} and collector potential V_C . (*Answer: $I_E = 2mA$, $V_{CE} = 8V$, $V_C = 18V$*)



VOLTAGE-DIVIDER BIAS

Review question

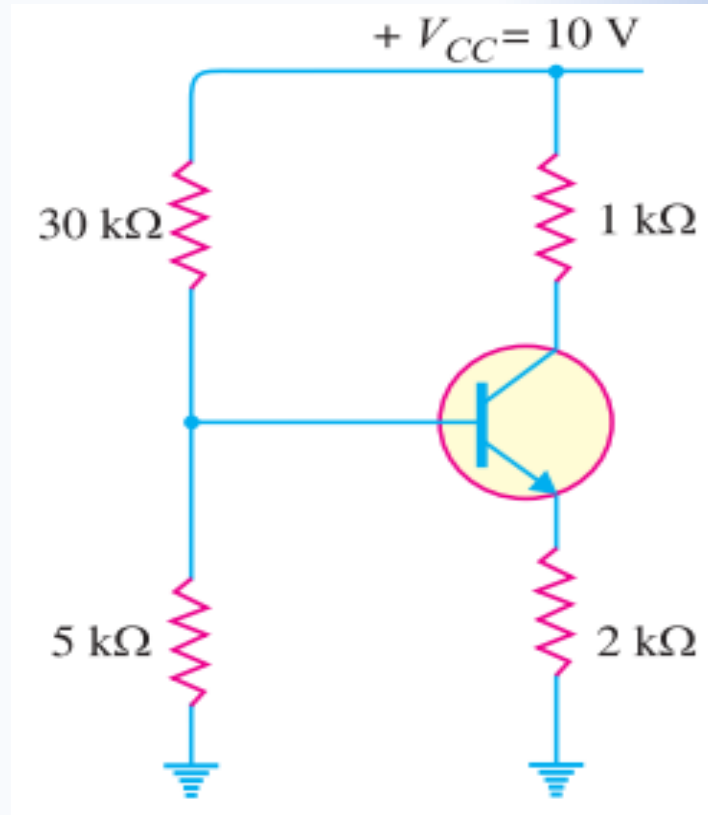
2. In the transistor circuit shown in fig below, find the operating point and draw dc load line. Assume the transistor to be of silicon. (*Answer:* $I_C = 10.5 \text{ mA}$, $V_{CE} = 7.75 \text{ V}$)



VOLTAGE-DIVIDER BIAS

Review question

3. In a transistor circuit shown in Fig below, find the operating point and draw dc load line. Assume silicon transistor is used. (*Answer: $I_C = 0.365 \text{ mA}$, $V_{CE} = 8.9 \text{ V}$*)



VOLTAGE-DIVIDER BIAS

Review question

4. Find the value of I_C for potential divider method if $V_{CC} = 9\text{ V}$, $R_E = 1\text{ k}\Omega$, $R_1 = 39\text{ k}\Omega$, $R_2 = 10\text{ k}\Omega$, $R_C = 2.7\text{ k}\Omega$, $V_{BE} = 0.15\text{ V}$ and $\beta = 90$. (*Answer: $I_C = 1.5\text{ mA}$*)



POWER RATING OF TRANSISTOR

- The maximum power that a transistor can handle without destruction is known as **power rating of the transistor**.
- When a transistor is in operation, almost all the power is dissipated at the reverse biased collector-base junction.
- The power rating (or maximum power dissipation) is given by :

$$\begin{aligned} P_{D(max)} &= \text{Collector current} \times \text{Collector base voltage} \\ &= I_C \times V_{CB} \end{aligned}$$

$$V_{CE} = V_{CB} + V_{BE}, \text{ since } V_{BE} \ll V_{CB} \text{ then } V_{CE} \simeq V_{CB}$$

$$P_{D(max)} = I_C \times V_{CE}$$



POWER RATING OF TRANSISTOR

- A transistor that is suitable for large power amplification is called a **power transistor**.
- It differs from other transistors mostly in size ; it is considerably larger to provide for handling the great amount of power.
- While connecting transistor in a circuit, it should be ensured that its power rating is not exceeded otherwise the transistor may be destroyed due to excessive heat.



POWER RATING OF TRANSISTOR

- For example, suppose the power rating (or maximum power dissipation) of a transistor is 300 mW.
- If the collector current is 30 mA, then maximum V_{CE} allowed is given by ;

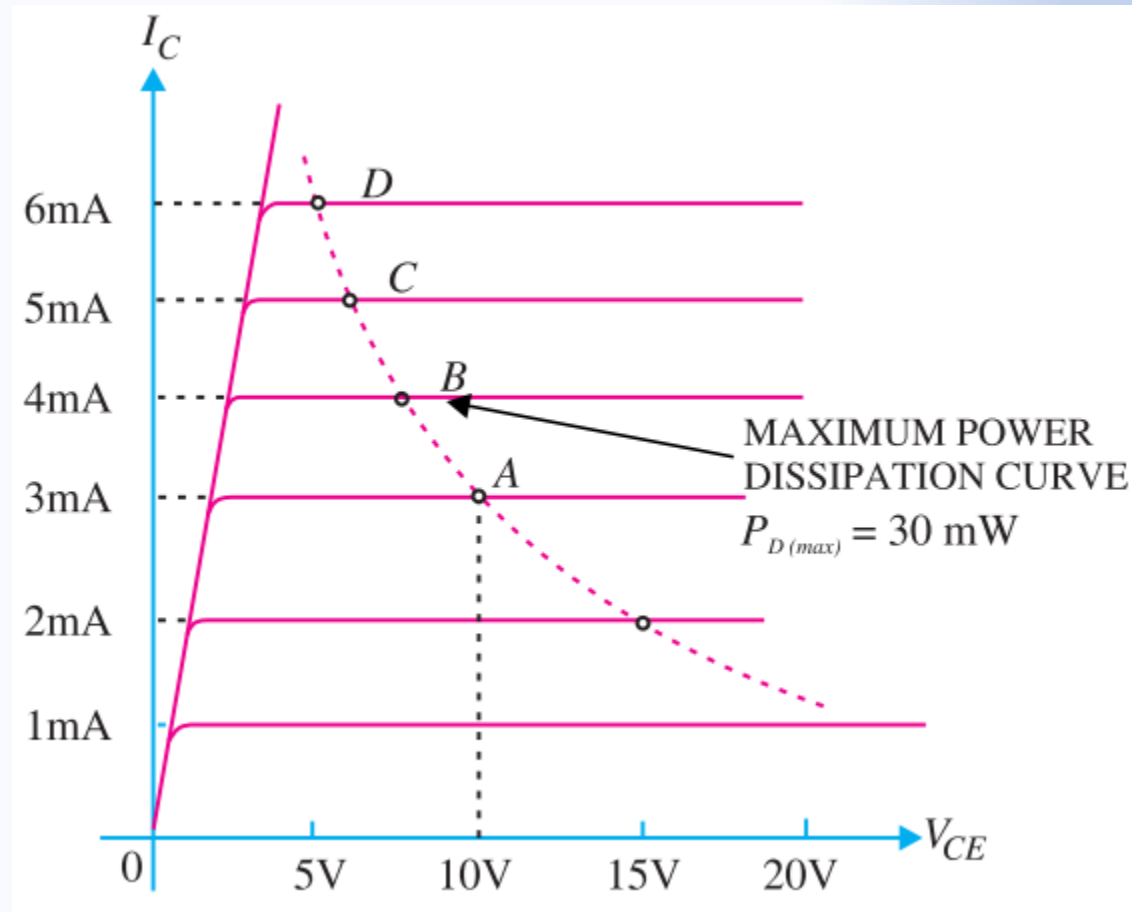
$$V_{CE (max)} = \frac{P_{D (max)}}{I_C} = \frac{300 \text{ mW}}{30 \text{ mA}} = 10V$$

- This means that for $I_C = 30 \text{ mA}$, the maximum V_{CE} allowed is 10V.
- If V_{CE} exceeds this value, the transistor will be destroyed due to excessive heat.



POWER RATING OF TRANSISTOR

- Maximum power dissipation curve



POWER RATING OF TRANSISTOR

Example 3

- For the circuit shown in Fig below, find the transistor power dissipation. Assume that $\beta = 200$.

Solution

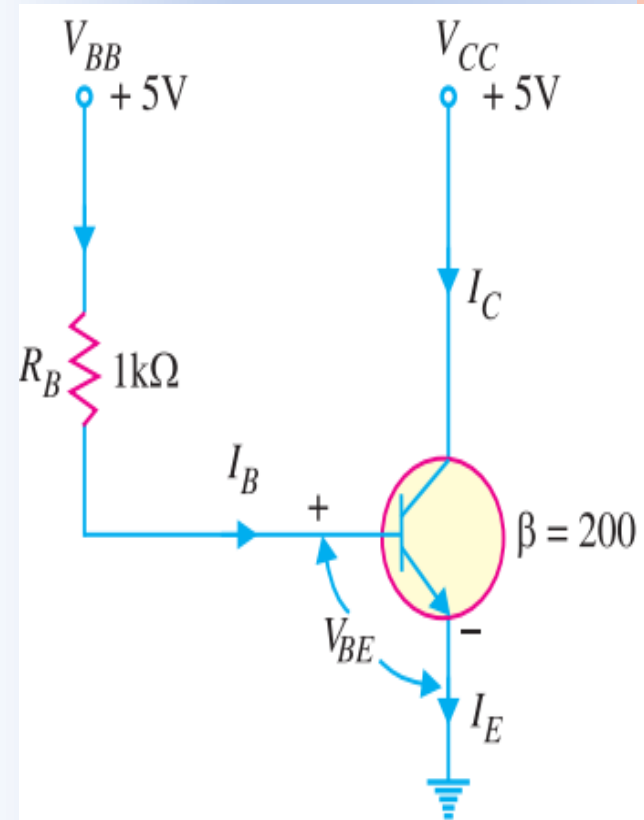
$$I_B = \frac{V_{BB} - V_{BE}}{R_B} = \frac{(5 - 0.7)V}{1 \text{ k}\Omega} = 4.3 \text{ mA}$$

$$I_C = \beta I_B = 200 \times 4.3 \text{ mA} = 860 \text{ mA}$$

$$V_{CE} = V_{CC} = 5 \text{ V}$$

Power dissipation

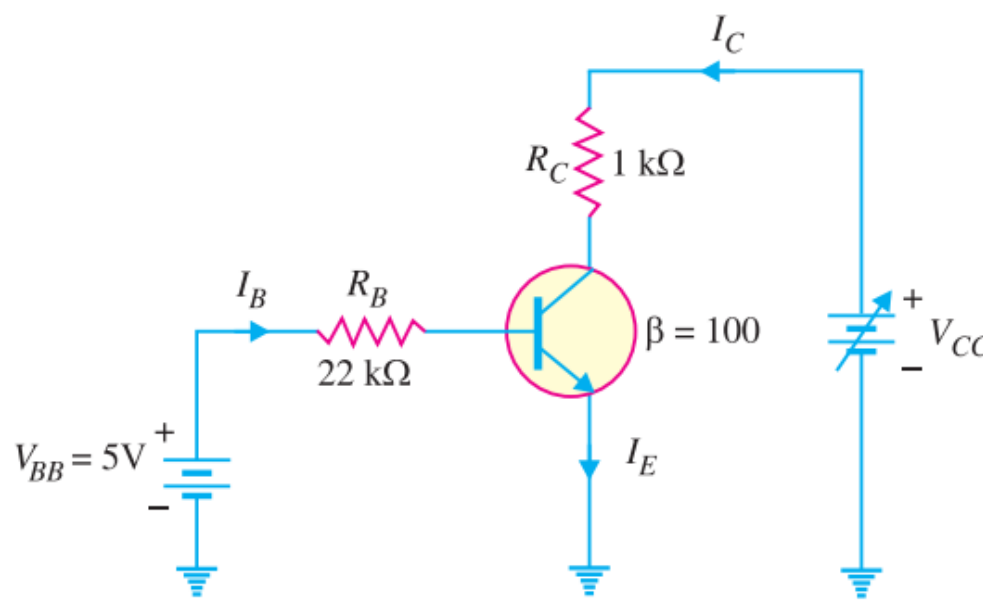
$$P_D = I_C \times V_{CE} = 860 \text{ mA} \times 5 \text{ V} = 4.3 \text{ W}$$



POWER RATING OF TRANSISTOR

Example 4

- The transistor in Fig below has the following maximum ratings: $P_{D(max)} = 800 \text{ mW}$; $V_{CE(max)} = 15\text{V}$; $I_{C(max)} = 100 \text{ mA}$. Determine the maximum value to which V_{CC} can be adjusted without exceeding any rating. Which rating would be exceeded first ?



POWER RATING OF TRANSISTOR

Solution

$$I_B = \frac{V_{BB} - V_{BE}}{R_B} = \frac{(5 - 0.7)V}{22 \text{ k}\Omega} = 195 \mu A$$

$$I_C = \beta I_B = 100 \times 195 \mu A = 19.5 \text{ mA}$$

- Note that I_C is much less than $I_{C(max)}$ and will not change with V_{CC} .
- It is determined only by I_B and β .
- Therefore, current rating is not exceeded.



POWER RATING OF TRANSISTOR

Solution

- Now, $V_{CC} = V_{CE} + I_C R_C$
- We can find the value of V_{CC} when $V_{CE (max)} = 15V$.

$$V_{CC (max)} = V_{CE (max)} + I_C R_C = 15 V + 19.5 mA \times 1 k = 34.5V$$

- Therefore, we can increase V_{CC} to $34.5V$ before $V_{CE (max)}$ is reached.



POWER RATING OF TRANSISTOR

Solution

$$P_D = I_C \times V_{CE (max)} = 19.5 \text{ mA} \times 15 \text{ V} = 293 \text{ mW}$$

- Since $P_D (max) = 800 \text{ mW}$, it is not exceeded when $V_{CC} = 34.5\text{V}$.
- If base current is removed causing the transistor to turn off, $V_{CE (max)}$ will be exceeded because the entire supply voltage V_{CC} will be dropped across the transistor.



TUTORIAL QUESTIONS

Book: Electronic Devices (Conventional Current Version) (9th Edition) by Thomas L. Floyd

Section 5 –1: Qn 1 to Qn 10

Section 5 –2: Qn 16 to Qn 18

Section 5 –4: Qn 33 to Qn 37



NEXT SESSION...DIGITAL ELECTRONICS

